

Binary Addition

By the end of this worksheet you should:

- know how a computer adds two binary values together
- be able to define what is meant by a **carry**



Computers can't do much; but they do it incredibly quickly.

One things computers can do is add.

Yet computers use binary, so we need to know how computers add two binary numbers together.



BINARY: A numbering system used by computers and digital circuits.
Uses 2 values; 0 or 1.

DENARY: A numbering system used by humans around the world.
Uses 10 values: 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9.

Why? Why do we need to know how to add two binary numbers together?

Well, it's not to check that you can add two 1s together; we're sure you can do that.

It's because knowing how your computer works is key to understanding programming and how computers do the things they do.

Now, it is important to remember the following:

$$\begin{aligned}0 + 0 &= 0 \\0 + 1 &= 1 \\1 + 0 &= 1 \\1 + 1 &= 2 \\1 + 1 + 1 &= 3\end{aligned}$$



Binary Addition

Again, this is obvious but we also know that computers only use binary (zeros and ones), so there is no 2 or 3.

Instead, we need to use the binary equivalent values for 2 and 3, which are:



0	00
1	01
2	10
3	11

And then we need to remember the old carry we do in our own mathematics.

Carry?

Consider adding these two denary values together:


$$\begin{array}{r} 2 \quad 5 \quad 1 \\ 1 \quad 7 \quad 5 \\ \hline 4 \quad 2 \quad 6 \\ 1 \end{array}$$

Look at the small 1 at the bottom of the last column; that is a carry. This is there because when we add 5 and 7 together we get a value bigger than 9; so, we need to carry the tens part of the answer into the next column.



CARRY: to transfer to the next column.

In maths, if the result of a calculation exceeds the maximum value for that column, the additional amount is 'carried over' to the next, more significant, column.

We do the same in binary:

$$\begin{array}{r} 0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \\ \hline 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ 1 \quad 1 \quad 1 \quad \quad \quad \quad 1 \end{array}$$



This takes a little getting used to but basically, $1 + 1 = 2$, which is 10 in binary. We place the 0 in the answer area and carry the 1 into the next column.

If we add $1 + 1 + 1$ (with the last 1 being a carry), then we end up with 11, so 1 goes into the answer area and we carry the 1 into the next column.

What if the carry continues beyond our 8 columns?

Well, the computer remembers this, putting in to a special area called the **Carry Flag**.



CARRY FLAG: A special register bit in the processor that is set to 1 if any addition results in a carry beyond the 8th bit.

What happens if we want to add 4 ones?

We never will.

Computers only ever add two binary numbers together, so the maximum value that can ever be achieved is 3; never 4.



It's always worth checking your answers by converting everything to denary first.

But remember, the examiner wants to see the carries; they want to see you know how a computer adds binary numbers together.

Try these...



a)

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \\ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \\ \hline \hline \end{array}$$

b)

$$\begin{array}{r} 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \\ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \\ \hline \hline \end{array}$$

c)

$$\begin{array}{r} 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \\ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1 \\ \hline \hline \end{array}$$

d)

$$\begin{array}{r} 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \\ \hline \hline \end{array}$$

e)

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \\ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \hline \hline \end{array}$$

f)

$$\begin{array}{r} 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \\ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \\ \hline \hline \end{array}$$